

HITIK PANCHAL

M.Sc. Computer Science, Universität Bonn | Köln, Germany

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Available as Working Student / Graduate / Entry-Level roles in AI / ML / NLP Engineering,
Data Analytics, or Software Engineering.



SUMMARY

Computer Science Master's student at Universität Bonn (Grade: 1.7) with 1.5 years of production experience building high-throughput backend systems at NSE India. Hands-on professional experience with Python, Java/Spring Boot, SQL, and REST APIs across large-scale financial systems. University research focused on NLP and LLMs - including LoRA fine-tuning of transformer models and multilingual narrative classification using RoBERTa and LLaMA. Personal projects spanning AI agents, prompt engineering, and analytics dashboards.

PROFESSIONAL EXPERIENCE

Associate System Analyst · NSEIT Private Limited

May 2023 – Feb 2024 · Mumbai, IN

- Architected and delivered migration of NSE Clearing Management System (NCMS) from Java/Spring Boot monolith to REST API-based architecture, increasing throughput by ~20% and enabling the platform to sustain **20,000+ trades/minute** for live clearing operations.
- Identified and resolved critical backend performance bottlenecks; DB query refactoring reduced database load by ~10% and eliminated several production bugs, improving overall system stability by **4-5%**.
- Served as project lead for NCMS CM module - owned delivery planning, stakeholder communication, and release coordination across a cross-functional team of engineers and QA analysts.
- Mentored 2 junior developers through structured code reviews, improving code quality standards and reducing review cycle time across the NCMS module.

Stack: Java, Spring Boot, MySQL, JavaScript, Linux, REST APIs, Postman, Git

Trainee Associate System Analyst · NSEIT Private Limited

Nov 2022 – May 2023 · Mumbai, IN

- Executed production deployments on Linux for NCMS serving **100,000+ end users**, maintaining near-zero downtime across all release cycles.
- Led Git migration of 10+ legacy NCMS codebases, standardising version control across the team and significantly reducing branch conflicts and manual merge overhead.
- Resolved critical SQL and application bugs during UAT and testing phases, reducing defect backlog and accelerating time-to-production for key features.

Stack: Java, Spring Boot, SQL, JavaScript, Linux, Git

Data Science Intern · Dezignolics Web and Software Company

Aug 2022 – Oct 2022 · Navi Mumbai, IN

- Built an end-to-end customer churn prediction pipeline (logistic regression + decision tree ensemble, scikit-learn) on **200,000+ subscription records**; optimised for AUC score and delivered risk scores surfaced via Tableau dashboards adopted by the marketing team for retention targeting.
- Designed and implemented ETL pipelines including data cleaning, feature engineering, and model tuning; Tableau dashboards were used directly in marketing retention strategy decisions.

Stack: Python, scikit-learn, TensorFlow, Tableau, SQL, ETL pipelines

EDUCATION

M.Sc. Computer Science · Universität Bonn

Apr 2024 – Present · Bonn, DE

Current Grade: 1.7 · Focus: Machine Learning, NLP, Large Language Models

B.E. Computer Engineering · Mumbai University

Jun 2018 – Mar 2022 · Navi Mumbai, IN

CGPA: 9.76 / 10 · Final-year thesis on real-time crowd density estimation using CNN/MCNN (~90% accuracy)

PERSONAL PROJECTS

26/27 Football Season Simulator *React · TypeScript · Zustand · Web Workers · Tailwind*

- Built a full browser-based football management simulator ([26-27-football-season-simulator.vercel.app](#)) modeling transfers, squad building, and full 38-match seasons for 12 European clubs (~3,500 real players) using a Dixon-Coles + Monte Carlo match engine and a full domestic/European cup engine, all running client-side on seeded PRNGs.

Got You CoVered - AI Cover Letter Generator *Next.js · TypeScript · Anthropic Claude API · Tailwind · Vercel*

- Built and deployed a **live** full-stack web app ([got-you-covered-cv.vercel.app](#)) that generates tailored one-page cover letters from a user's resume and a target job description using the Anthropic Claude API - supporting custom instructions, an optional reference cover letter to emulate the user's voice, tone/length controls, on-the-spot editing of the generated letter and one-click .docx/.pdf export, with all user data kept client-side.

Value Alignment via Domain-Specific LoRA Fine-Tuning *Python · PyTorch · Qwen3 · LoRA/PEFT · Hugging Face*

- Researched the effect of domain-specific pretraining on value orientations of Qwen3 models (0.6B-8B) using LoRA fine-tuning on five subsets of The Pile; evaluated value shifts against the Kiesel Value Survey (Schwartz theory). Found that shifts in smaller models were driven primarily by output homogenisation rather than substantive value acquisition - a finding with direct implications for responsible LLM deployment.

Multilingual Narrative Classification in News Articles *Python · PyTorch · RoBERTa · LLaMA-3.1-8B · Hugging Face*

- Built a propaganda and narrative detection system for English/Russian news using a dual-head RoBERTa multi-label classifier with LLaMA-3.1-8B-Instruct refinement for low-resource classes. Achieved **85% multi-label accuracy** on a highly imbalanced dataset - approximately 18pp above baseline - enabling automated monitoring of disinformation narratives at scale.

AI-Assisted Agile Planning & Analytics Dashboard *Python · Pandas · Streamlit · REST APIs · Prompt Engineering*

- Designed and prototyped a dashboard to analyse Jira-style sprint data (epics, capacity, dependencies, delivery risks) and generate AI-assisted planning summaries, backlog refinement inputs, and leadership reports - converting raw project data into structured decision support for engineering leads.

Financial Portfolio Analysis Dashboard *Tableau · Python · Yahoo Finance API · Excel*

- Designed a Tableau dashboard comparing portfolio performance across equities, bonds, and ETFs using historical Yahoo Finance data; integrated dynamic filters and comparative charts enabling side-by-side risk/return analysis - demonstrating applied financial analytics and BI skills directly relevant to investment or fintech contexts.

TECHNICAL SKILLS

Languages	Python, Java, JavaScript, C/C++, Shell, SQL, R
ML / NLP / AI	PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, RoBERTa, LLaMA, Qwen3, LoRA (PEFT), LangChain, Prompt Engineering
Data & Viz	Pandas, NumPy, Tableau, Power BI, Google Data Studio, Streamlit, Excel
Backend & APIs	Spring Boot, REST APIs, MySQL, MongoDB, Firebase, Docker, Kubernetes (familiar)
Tools & DevOps	Git, Jira, Linux, Postman, Confluence, LaTeX, Microsoft Office
Agile / PM Analytics	Sprint Planning, KPI Tracking, Capacity Analysis, Backlog Refinement, Dependency Tracking

CERTIFICATIONS

Google Data Analytics Professional Certificate (Coursera) · IBM Data Science Professional Certificate (Coursera)

Google Business Intelligence Professional Certificate (Coursera) · Data Visualization with Tableau (UC Davis, Coursera)

ADDITIONAL INFORMATION

Languages: English (Fluent, IELTS 7.5) · Hindi (Fluent) · German (A2, actively improving)

Interests: Football analytics · Chess · Photography